

# Linear Regression

## Advanced Statistical Inference

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### Linear regression

#### Objectives for today

1. Review of linear regression
2. Understand the probabilistic interpretation of (regularized) loss minimization

*Break*

3. Introduction of Bayesian linear regression
4. Compute the posterior distribution and make predictions
5. Model selection and other properties of Bayesian linear regression

*Break*

6. Class exercise on Bayesian inference for coin toss

### A quick recap on probability

Consider two continuous random variables  $x$  and  $y$

- Sum rule (marginalization):

$$p(x) = \int p(x, y) dy$$

- Product rule (conditioning):

$$p(x, y) = p(x | y)p(y) = p(y | x)p(x)$$

- Bayes' rule:

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)}$$


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Consider a random vector  $\mathbf{x}$  with  $D$  components ( $\mathbf{z} \in \mathbb{R}^D$ )

- Chain rule:

$$p(\mathbf{z}) = p(z_1, z_2, \dots, z_D) = p(z_1)p(z_2 \mid z_1)p(z_3 \mid z_1, z_2) \cdots p(z_D \mid z_1, z_2, \dots, z_{D-1})$$

If  $z_i$  are independent, then

$$p(z_1 \mid z_2, \dots, z_{D-1}) = p(z_1)$$

and the chain rule becomes

$$p(\mathbf{z}) = p(z_1)p(z_2) \cdots p(z_D) = \prod_{d=1}^D p(z_d)$$

## Definitions

- **Input**, features, covariates:  $\mathbf{x} \in \mathbb{R}^D$

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} \in \mathbb{R}^{N \times D} \quad \text{or with a bias term} \quad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x}_1 \\ \vdots & \vdots \\ \mathbf{1} & \mathbf{x}_N \end{bmatrix} \in \mathbb{R}^{N \times (D+1)}$$

- **Output**, target, response:  $y \in \mathbb{R}$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N$$

- **Dataset**:  $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$

## Regression

- **Objective:** Learn a function  $f: \mathbb{R}^D \rightarrow \mathbb{R}$

**Linear models** implement a linear combination of (basis) functions

$$f(\mathbf{x}) = \sum_{d=1}^D w_d \varphi_d(\mathbf{x}) = \mathbf{w}^\top \boldsymbol{\varphi}(\mathbf{x})$$

- **Parameters:**  $\mathbf{w} = [w_1, \dots, w_D]^\top$
- **Basis functions:**  $\boldsymbol{\varphi}(\mathbf{x}) = [\varphi_1(\mathbf{x}), \dots, \varphi_D(\mathbf{x})]^\top$

! Important

Any model that can be written as a linear combination of parameters (**not** the input) is a linear model

## Linear regression

For simplicity, let's consider linear functions

$$f(\mathbf{w}, \mathbf{x}) = \sum_{d=1}^D w_d x_d = \mathbf{w}^\top \mathbf{x}$$

- **Objective:** Find  $\mathbf{w}$  that minimizes the error

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y_n - f(\mathbf{w}, \mathbf{x}_n))^2 = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

## Least squares solution

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

- **Gradient:**  $\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}) = -\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) = 0$
- **Solution:**  $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$

### 💡 Exercise

Implement the least squares solution for linear regression using Cholesky decomposition and back-substitution (ref [revision of linear algebra](#))

## Probabilistic interpretation of linear regression

Minimizing the loss is equivalent to maximizing the likelihood of the data under a Gaussian noise model

$$\exp(-\gamma \mathcal{L}_i) \propto \mathcal{N}(\mathbf{y} \mid \mathbf{X}\mathbf{w}, \gamma^{-1}\mathbf{I})$$

**(Implicit) Assumption:** Data is generated by a linear model with **Gaussian noise**  $\varepsilon \sim \mathcal{N}(0, \sigma^2)$  independent across samples (with  $\sigma^2 = \gamma^{-1}$ ).

$$y_n = \mathbf{w}^\top \mathbf{x}_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2)$$

## Maximum likelihood estimation

Maximum likelihood estimation is solving

$$\arg \max_{\mathbf{w}} \prod_{i=1}^N \underbrace{\mathcal{N}(\mathbf{y}_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)}_{p(\mathbf{y}_i \mid \mathbf{w}, \mathbf{x}_i)}$$

$p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}) = \prod_{i=1}^N p(\mathbf{y}_i \mid \mathbf{w}, \mathbf{x}_i)$  is the **likelihood** of the data given the model

### 💡 Tip

We *never* maximize the likelihood directly, but the log-likelihood

$$\arg \max_{\mathbf{w}} \sum_{i=1}^N \log \mathcal{N}(\mathbf{y}_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)$$

Because log is monotonic and concave, the optimum value is the same and numerically more stable

## Likelihood is not a probability

- The likelihood function is not a probability distribution.
- The likelihood can take any non-negative value, not just values between 0 and 1.
- It represents the density of the data ( $\mathbf{y}$ ) given the model ( $\mathbf{w}$ ).

### Key insight:

- **Probability:** Fix the parameters, vary the data. It answers “what data might we see?”
- **Likelihood:** Fix the data, vary the parameters. It answers “which parameters best explain the data?”

## Properties of maximum likelihood estimation

- **Consistency:** As  $N \rightarrow \infty$ , the MLE converges to the true parameter value

### ⚠ Note

The consistency property makes sense if the model is correct (e.g. the data is generated by a linear model with Gaussian noise). But this is an assumption that is often not met in practice.

- **Unbiasedness:** The expected value of the MLE is *unbiased*

$$\mathbb{E}_{p(\mathbf{y}|\mathbf{w},\mathbf{X})}[\hat{\mathbf{w}}] = \mathbf{w}$$

## Proof that MLE is unbiased

The proof is quite easy

$$\begin{aligned}\mathbb{E}_{p(\mathbf{y}|\mathbf{w},\mathbf{X})}[\hat{\mathbf{w}}] &= \int \hat{\mathbf{w}} p(\mathbf{y} | \mathbf{w}, \mathbf{X}) d\mathbf{y} \\ &= \int (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \mathcal{N}(\mathbf{y} | \mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}) d\mathbf{y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \int \mathbf{y} \mathcal{N}(\mathbf{y} | \mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}) d\mathbf{y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \mathbf{w} = \mathbf{w}\end{aligned}$$

## Regularization

**Ridge regression** adds a penalty term to the loss function

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

- **Objective:** Find  $\mathbf{w}$  that minimizes the error while keeping the parameters small
- **Solution:**  $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$

## Probabilistic interpretation of regularization

- **Assumption:** Data is generated by a linear model with Gaussian noise independent across samples

Same trick as before (exponential of the negative loss)

$$\begin{aligned} \exp(-\gamma \mathcal{L}) &= \exp\left(-\frac{\gamma}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 - \frac{\gamma}{2} \lambda \|\mathbf{w}\|_2^2\right) = \\ &= \exp\left(-\frac{\gamma}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2\right) \exp\left(-\frac{\gamma}{2} \lambda \|\mathbf{w}\|_2^2\right) = \\ &\propto \mathcal{N}(\mathbf{y} \mid \mathbf{X}\mathbf{w}, \gamma^{-1} \mathbf{I}) \mathcal{N}(\mathbf{w} \mid \mathbf{0}, (\gamma\lambda)^{-1} \mathbf{I}) \end{aligned}$$

Minimizing the loss is equivalent to maximizing the product of two Gaussian distributions (likelihood and prior).

We are getting closer to Bayesian inference!

## Bayesian linear regression

### Bayesian inference

**Bayesian inference** allows to “transform” a prior distribution over the parameters into a posterior **after** observing the data

Bayes’ rule :

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}) p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}$$

- **Prior:**  $p(\mathbf{w})$

- Encodes our beliefs about the parameters **before** observing the data
- **Likelihood:**  $p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})$ 
  - Encodes our model of the data
- **Posterior:**  $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$ 
  - Encodes our beliefs about the parameters **after** observing the data (e.g. conditioned on the data)
- **Evidence:**  $p(\mathbf{y} \mid \mathbf{X})$ 
  - Normalizing constant, ensures that  $\int p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) d\mathbf{w} = 1$

## Bayesian linear regression - Likelihood and prior

Modeling observation as noisy realization of a linear combination of the features As before, we assume a Gaussian likelihood

$$p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}) = \mathcal{N}(\mathbf{y} \mid \mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I})$$

For the prior, we use a Gaussian distribution over the model parameters

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \mathbf{S})$$

In practice, we often use a diagonal covariance matrix  $\mathbf{S} = \sigma_w^2 \mathbf{I}$

## When can we compute the posterior?

### Definition

A prior is **conjugate** to a likelihood if the posterior is in the same family as the prior.

Only a few conjugate priors exist, but they are very useful.

Examples:

- Gaussian likelihood and Gaussian prior  $\Rightarrow$  Gaussian posterior
- Binomial likelihood and Beta prior  $\Rightarrow$  Beta posterior

Full table available on [wikipedia](#)

## Why is this useful?

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}$$

- **Generally** the posterior is **intractable** to compute
  - We don't know the form of the posterior  $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$
  - The evidence  $p(\mathbf{y} \mid \mathbf{X})$  is an integral
    - \* without closed form solution
    - \* high-dimensional and computationally intractable to compute numerically

...

- **Analytical solution** thanks to conjugacy:
  - We know the form of the posterior
  - We know the form of the normalization constant
  - We don't need to compute the evidence, just some algebra to get the posterior

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Back to our model, the posterior must be Gaussian  $\mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$

Ignoring constant terms in  $\mathbf{w}$ :

$$\begin{aligned} p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) &\propto \exp\left(-\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{w} - \boldsymbol{\mu})\right) \\ &= \exp\left(-\frac{1}{2}\left(\mathbf{w}^\top \boldsymbol{\Sigma}^{-1} \mathbf{w} - 2\mathbf{w}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} + \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}\right)\right) \\ &\propto \exp\left(-\frac{1}{2}\left(\mathbf{w}^\top \boldsymbol{\Sigma}^{-1} \mathbf{w} - 2\mathbf{w}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}\right)\right) \end{aligned}$$

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From the likelihood and prior, we can write the posterior as

$$\begin{aligned} p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w}) &\propto \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 - \frac{1}{2}\mathbf{w}^\top \mathbf{S}^{-1}\mathbf{w}\right) \\ &\propto \exp\left(-\frac{1}{2}\left(\mathbf{w}^\top \left(\frac{1}{\sigma^2}\mathbf{X}^\top \mathbf{X} + \mathbf{S}^{-1}\right)\mathbf{w} - \frac{2}{\sigma^2}\mathbf{w}^\top \mathbf{X}^\top \mathbf{y}\right)\right) \end{aligned}$$

...

From previous slide,



$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) \propto \exp \left( -\frac{1}{2} \left( \mathbf{w}^\top \boldsymbol{\Sigma}^{-1} \mathbf{w} - 2 \mathbf{w}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \right) \right)$$

We can identify the posterior mean and covariance

**Posterior covariance**

$$\boldsymbol{\Sigma} = \left( \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \mathbf{S}^{-1} \right)^{-1}$$

**Posterior mean**

$$\boldsymbol{\mu} = \frac{1}{\sigma^2} \boldsymbol{\Sigma} \mathbf{X}^\top \mathbf{y}$$

## How to make predictions?

The posterior distribution  $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$  gives us the uncertainty about the parameters. **How can we use it to make predictions?**

...

The predictive distribution is the distribution of the target variable  $y_\star$  given the input  $\mathbf{x}_\star$

Obtained by **marginalizing** the parameters using the posterior

$$p(y_\star \mid \mathbf{x}_\star, \mathbf{y}, \mathbf{X}) = \int p(y_\star \mid \mathbf{w}, \mathbf{x}_\star) p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) d\mathbf{w}$$

...

For linear regression, the predictive distribution is Gaussian

$$p(y_\star \mid \mathbf{x}_\star, \mathbf{y}, \mathbf{X}) = \mathcal{N}(y_\star \mid \boldsymbol{\mu}^\top \mathbf{x}_\star, \mathbf{x}_\star^\top \boldsymbol{\Sigma} \mathbf{x}_\star + \sigma^2)$$

## Bayesian linear regression with basis functions

The same approach can be used with non-linear basis functions

Transform the input  $\mathbf{x}$  using a non-linear function  $\varphi(\mathbf{x})$

$$\mathbf{x} \rightarrow \varphi(\mathbf{x}) = [\varphi_1(\mathbf{x}), \dots, \varphi_D(\mathbf{x})]^\top$$

For convenience, define  $\Phi = [\varphi(\mathbf{x}_1), \dots, \varphi(\mathbf{x}_N)]^\top$

**Posterior**

$$p(\mathbf{w} \mid \mathbf{y}, \Phi) = \mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}, \Sigma) \quad \text{with} \quad \boldsymbol{\mu} = \frac{1}{\sigma^2} \Sigma \Phi^\top \mathbf{y} \quad \text{and} \quad \Sigma = \left( \frac{1}{\sigma^2} \Phi^\top \Phi + \mathbf{S}^{-1} \right)^{-1}$$

**Predictive distribution**

$$p(y_\star \mid \mathbf{x}_\star, \mathbf{y}, \Phi) = \mathcal{N}(y_\star \mid \boldsymbol{\mu}^\top \varphi(\mathbf{x}_\star), \varphi(\mathbf{x}_\star)^\top \Sigma \varphi(\mathbf{x}_\star) + \sigma^2)$$

Where we used polynomial basis functions  $\varphi_i(\mathbf{x}) = x^i$ :  $f(\mathbf{w}, \mathbf{x}) = \sum_{i=0}^K w_i x^i$

## Analysis of Bayesian linear regression

### Connection with ridge regression and maximum a posteriori estimation

**Maximum a posteriori estimation** (MAP) computes the mode of the posterior distribution  $\arg \max p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$ . For Gaussians, it is the same as the mean

$$\arg \min \mathcal{L}(\mathbf{w}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

$$\arg \max p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) = \mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}, \Sigma)$$

If  $\lambda = \sigma^2 / \sigma_w^2$ , the **ridge regression** solution is equivalent to the MAP solution with a Gaussian prior

## Effect of the prior

- Prior encodes our beliefs about the parameters before observing the data.
- Prior effect diminishes with more data
- When we don't have much data, the prior can have a strong effect on the posterior

...

**Question:** How do we choose the prior?

1. Data type:

- Real-values: Gaussian prior
- Positive data: Log-normal prior, Gamma prior, etc.
- 0-1 data: Beta prior
- Data summing to 1: Dirichlet prior

2. Expert knowledge

3. Computational convenience

## Model selection

We can now get a solution for the linear regression problem (Bayesian and not) but we have to choose the model

**Questions:**

- What is the best model for the data?
- How many basis functions should we use?
- How to avoid overfitting?

Attempted to choose the model based on the likelihood of the data. Is this a good idea?

**NO!**

- Higher complexity models will always have higher likelihood ...
- ... but they will also overfit the data and generalize poorly

## Model selection - Cross-validation

- **Cross-validation:** Split the data into training and validation sets
- Solve the model with the training set and evaluate the performance on the validation set
- (Optional) Repeat the process for different splits of the data
- Choose the model that performs best on the validation set

### 💡 Pros

- Simple
- Works well in practice

### ❗ Cons

- Computationally expensive
- Requires multiple runs
- Works poorly with small datasets
- **Violates the likelihood principle**

## Likelihood principle

- **Likelihood principle:** All the information from the data is contained in the likelihood function
- **Consequence:** You should *not* base your inference on data that you *could have* observed but did not
- **Cross-validation** (and other frequentist methods) violate the likelihood principle

### 💡 Note

- This is more of a philosophical point than a practical one.
- It's not a *rule of nature* but an argument that Bayesian methods are more *principled*.

## Marginal likelihood

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}$$

The **marginal likelihood** is the normalization constant of the posterior distribution

$$p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w}) d\mathbf{w}$$

- We are averaging the likelihood over **all possible values** of the parameters from the prior
  - It tells us how likely the data is under the model
- 

Let's be explicit about the model in the marginal likelihood

$$p(\mathbf{y} \mid \mathbf{X}, \mathcal{M}) = \int p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}, \mathcal{M})p(\mathbf{w} \mid \mathcal{M}) d\mathbf{w}$$

where  $\mathcal{M}$  is the model (e.g. polynomial degree) and  $p(\mathbf{w} \mid \mathcal{M})$  is the prior over the parameters for the model  $\mathcal{M}$ .

- Recipe for **Bayesian model selection**:

$$\widehat{\mathcal{M}} = \arg \max_{\mathcal{M}} p(\mathbf{y} \mid \mathbf{X}, \mathcal{M})$$

This is also known as **Type II maximum likelihood** or **evidence maximization**

## Model selection with Bayesian linear regression

Given:

- *Likelihood*:  $p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}) = \mathcal{N}(\mathbf{y} \mid \mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I})$
- *Prior*:  $p(\mathbf{w} \mid m) = \mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$

The **marginal likelihood** is a Gaussian

$$p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{y} \mid \mathbf{X}\boldsymbol{\mu}_p, \mathbf{X}\boldsymbol{\Sigma}_p\mathbf{X}^\top + \sigma^2 \mathbf{I})$$

It does NOT depend on the parameters  $\mathbf{w}$  but only on the model!

## Why does it work? The Bayesian Occam's razor

Does  $p(\mathbf{y} \mid \mathbf{X}, \mathcal{M})$  favor complex models? **No!**

- We *marginalize* over the parameters, not *maximize* them
- The marginal likelihood penalizes complex models that don't fit the data well

This is known as the **Bayesian Occam's razor**: the simplest model that explains the data is the best

### 💡 Intuition

Apply chain rule to the marginal likelihood (drop all dependencies):

$$p(\mathbf{y}) = p(y_1)p(y_2 \mid y_1)p(y_3 \mid y_1, y_2) \dots p(y_N \mid y_1, \dots, y_{N-1})$$

or equivalently

$$\log p(\mathbf{y}) = \log p(y_1) + \log p(y_2 \mid y_1) + \log p(y_3 \mid y_1, y_2) + \dots + \log p(y_N \mid y_1, \dots, y_{N-1})$$

- If the model is too complex, it will predict early data points well but later data points poorly

## Estimating the noise variance

So far, we treated  $\sigma^2$  as known. Can we estimate it from data?

Remember: in our model assumption, the data is generated by a linear model with Gaussian noise

$$y_n = \mathbf{w}^\top \mathbf{x}_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2)$$

## Maximum likelihood estimate of $\sigma^2$

Using  $\hat{\mathbf{w}}_{\text{ML}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  and maximizing

$$\log p(\mathbf{y} \mid \hat{\mathbf{w}}_{\text{ML}}, \sigma^2, \mathbf{X})$$

we get

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{N} \|\mathbf{y} - \mathbf{X} \hat{\mathbf{w}}_{\text{ML}}\|_2^2$$

 Bias of the ML estimator

For finite  $N$ , this estimator is biased downward:

$$\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \frac{N-D}{N}\sigma^2 = \sigma^2 - \frac{D}{N}\sigma^2$$

because  $D$  degrees of freedom are used to estimate  $\mathbf{w}$ .

## A more flexible noise model

- So far: same variance for all points
- But this is a **modeling assumption** that might not be realistic in many cases.
- How can we model more flexible noise?

...

**Heteroskedastic model:**

$$\varepsilon_n \sim \mathcal{N}(0, \sigma_n^2)$$

with  $\sigma_n^2$  that can vary across samples (e.g. as a function of the input  $\mathbf{x}_n$ )

Solving this model is more complex, unless we make additional assumptions on how  $\sigma_n^2$  varies across samples (e.g. linear function of the input).